Manchester Metropolitan University

Introduction to Number Theory

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Introduction to the unit

- Teaching team: Dr Killian O’Brien  and Dr Safiullah Khan  . See Moodle for contact details.
- 6G6Z0024 Applied Cryptography (15 credits)
- Assessment is 100% coursework. A portfolio of exercises.
- Timetable
- Let’s look at the Moodle page for the unit.

Introduction to Number Theory

We deal with the positive and negative *counting* numbers, more properly named the *integers*, and denoted by the symbol $\mathbb{Z}$, (coming from the German *Zahl*, for number)

- $\mathbb{Z} = \{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\}$
- $\mathbb{Z}$ is an *infinite* set.
- $\mathbb{Z}$ obviously carries the operations of addition, $+$, and multiplication, $\cdot$, that you've known from primary school.

Modern cryptography relies heavily on techniques and facts from *number theory*, which is the mathematical study of the integers and their properties under $+$ and $\cdot$.

Topics we need to know

- The **divisibility relation** on $\mathbb{Z}$.
- **Greatest Common Divisors** (gcd) and the **Euclidean Algorithm**.
- The **congruence relation** and **modular arithmetic**.
- **Prime** numbers and
    - The **Fundamental Theorem of Arithmetic** and **prime factorizations**
    - **Fermat's Little Theorem**
    - **Euler's totient** function
    - **Euler's theorem**
    - **Primality** testing, the **Miller-Rabin** test
- **The Chinese Remainder Theorem**
- **Discrete logarithms**

All these covered in Stallings, Chapter 2: Introduction to Number Theory.

The divisibility relation

- Recall, a *relation* in computer science / mathematics is a formula $A(x_1, \dots, x_n)$, so that when values are supplied for the variables $x_1, \dots, x_n$, results in a *statement* $A(x_1, \dots, x_n)$, i.e. something which is true or false.
- For a pair of integers a, b, with $b \neq 0$, we say b *divides* a, and write $b \mid a$ if there exists an integer c such that

$$a = b \cdot c,$$

  and if no such integer c exists then we say b does *not divide* a, and can write $b \nmid a$.
- So $b \mid a$ is a binary relation on a, b, i.e. a statement that is true or false, depending on the values of a, b.
- If $b \mid a$ then we say b is a *factor* or *divisor* of a.

Examples

- $3 \mid 15, 5 \mid 15, 1 \mid 15, 15 \mid 15.$
- $3 \nmid 10, 17 \nmid 20.$

Properties of divisibility

The divisibility relation enjoys the following properties, which can all be demonstrated (and proved) using its definition and basic properties of the integers.

- If $a|1$ then $a = \pm 1$, i.e. $a = -1$ or $a = +1$.
- If $a|b$ and $b|a$ then $a = \pm b$.
- For all non-zero integers b, we have $b|0$, i.e. *everything divides 0*.
- If $a|b$ and $b|c$ then $a|c$, i.e. the divisibility relation is *transitive*, it travels through intermediaries.
- If $x|y$ and $x|z$ then for all pairs of integer coefficients α, β, we have

$$x|(\alpha \cdot y + \beta \cdot z),$$

  i.e. x divides all *linear combinations* of y and z.

To familiarise yourself with these, work through some examples of the transitivity of divisibility and the divisibility of linear combinations.

The integer division algorithm

Do you remember this kind of thing from primary school?

- 20 divided by 3, goes in 6 times, with remainder 2.
- $20 = 6 \cdot 3 + 2$

The *integer division algorithm* is simply a formalization of this. It is:

- Given any positive integer n and any non-negative integer a, we can divide a by n to get an integer quotient q and remainder r that satisfy
- $a = qn + r$, and $0 \leq r < n$, and $q = \lfloor a/n \rfloor$
- $\lfloor x \rfloor$ is defined as the largest integer less than x, the so-called *floor* function.

Greatest Common Divisors (gcd)

We write $\gcd(a, b)$ for the *greatest common divisor of a and b*. So $\gcd$ is defined by

- $\gcd(a, b) = d$, where d is the largest integer that divides both a and b.
- For neatness, we also define $\gcd(0, 0) = 0$.

For example

- $\gcd(60, 24) = 12, \gcd(100, 75) = 25, \gcd(15, 32) = 1$.
- Note that, by its definition, $\gcd$ will always be non-negative, i.e. $\gcd(-60, 24) = 12$.

For small arguments a, b, we can calculate $\gcd(a, b)$ *in our heads*, so to speak.

- $\gcd(25, 3) =?, \gcd(99, 27) =?, \ldots.$
- But what about $\gcd(12349878973245, 324765)$?

The Euclidean Algorithm

In fact there is a classic algorithm that can quickly determine $\gcd$, and establishes the following, non-obvious fact,

- $\gcd(a, b)$ is the smallest positive integer d that can be written in the form

$$d = x \cdot a + y \cdot b,$$

for integer coefficients x, y.

The Euclidean algortihm was known to ancient mathematicians and has several important uses and generalisations in mathematics and cryptography.

The Euclidean Algorithm

A detailed treatment is given in Stallings. The algorithm depends on the following property of $\gcd$.

- If $a = qn + r$ then $\gcd(a, n) = \gcd(n, r)$.

This is true because

- if d is a common divisor of a and n, then since $r = a - qn$, i.e. r is a linear combination of a and n, then d divides r also. And so d is a common divisor of n and r.
- Similarly we can show that if e is a common divisor of n and r, then e divides a also. And so e will be a common divisor of a and n.
- So the pairs (a, n) and (n, r) have the exact same set of common divisors.
- Therefore,

$$\gcd(a, n) = \gcd(n, r).$$

The Euclidean Algorithm

The algorithm works by repeatedly applying the property from the last slide, to a sequence of integer divisions, until the $\gcd$ is clear. Best seen with a worked example

- What is $\gcd(710, 310)$?
- $710 = 2 \cdot 310 + 90$ so $\gcd(710, 310) = \gcd(310, 90)$,
- $310 = 3 \cdot 90 + 40$ so $\gcd(310, 90) = \gcd(90, 40)$,
- $90 = 2 \cdot 40 + 10$ so $\gcd(90, 40) = \gcd(40, 10)$,
- $40 = 4 \cdot 10 + 0$ so $\gcd(40, 10) = \gcd(10, 0) = 10$.

Note that

- The algorithm will terminate, since the remainders are a strictly decreasing sequence of non-negative integers.
- By definition of divisibility, $\gcd(x, 0) = x$, for all integers x.
- The $\gcd$ equations associated to the integer divisions all link together.
- So we can conclude that

$$\gcd(710, 310) = 10.$$

See Stallings for the full detail, a flowchart specification of the algorithm, and more examples.

The mod operator and the congruence relation

For an integer a and a positive integer n we say that a *modulo* n is the remainder r in the integer division of a by n.

- $a = qn + r, \quad 0 \leq r < n$
- We write $(a \mod n) = r$.
- n is called the *modulus* in this expression.

For example

- $(11 \mod 7) = 4$ and $(-11 \mod 4) = 1$.

There is an associated binary relation here.

We say that two integers a and b are *congruent modulo* n, written as

$$a \equiv b \pmod{n},$$

if

- $(a \mod n) = (b \mod n)$
- That is, if a and b *leave the same remainder*, after division by n.

The mod operator and the congruence relation

Examples

- $23 \equiv 8 \pmod{5}$
- $-11 \equiv 5 \pmod{8}$
- $81 \equiv 0 \pmod{27}$

The congruence relation has the following properties

- $a \equiv b \pmod{n}$ if and only if $n|(a-b)$
- $a \equiv a \pmod{n}$, called *reflexivity*
- $a \equiv b \pmod{n}$ implies that $b \equiv a \pmod{n}$, called *symmetry*
- If $a \equiv b \pmod{n}$ and $b \equiv c \pmod{n}$, then $a \equiv c \pmod{n}$, called *transitivity*
- These last three properties mean congruence modulo n is an *equivalence relation* on $\mathbb{Z}$.

Modular arithmetic

- The mod operator $(a \mod n)$ maps all integers a into the set

$$\mathbb{Z}_n = \{0, 1, 2, 3, \ldots, n-1\}.$$

- This is the set of *residues*, or *remainders*, modulo n.
- The familiar operations of $+$ and $\cdot$ on $\mathbb{Z}$ extend to $\mathbb{Z}_n$ in a natural way.

$$(a \mod n) + (b \mod n) := ((a+b) \mod n)$$

$$(a \mod n) \cdot (b \mod n) := ((a \cdot b) \mod n)$$

This means that $\mathbb{Z}_n$, with the operations of $+$ and $\cdot$ will form a *closed system* with respect to these operations, i.e. for any pair x, y from $\mathbb{Z}_n$, $x + y$ and $x \cdot y$ will again be elements of $\mathbb{Z}_n$.

See Stallings for worked examples of $\mathbb{Z}_8$ under $+$ and $\cdot$.

Modular arithmetic

- So given x from $\mathbb{Z}_n$, x will have an *additive inverse*, $n - x$, which satisfies

$$x + (n - x) \equiv 0 \pmod{n}.$$

- Given x from $\mathbb{Z}_n$, if there exists a y in $\mathbb{Z}_n$ which satisfies

$$x \cdot y \equiv 1 \pmod{n},$$

  then we say y is the *multiplicative inverse of x modulo n*, and vice versa. We can write $y \equiv x^{-1} \pmod{n}$.
- But multiplicative inverses do not necessarily exist for every element of $\mathbb{Z}_n$.

This is connected to the issue of cancellation in $\mathbb{Z}_n$.

- If $(a+b) \equiv (a+c) \pmod{n}$ then $b \equiv c \pmod{n}$.
- If $(a \cdot b) \equiv (a \cdot c) \pmod{n}$ then it's not necessarily true that $b \equiv c \pmod{n}$.
- However, if $a^{-1} \pmod{n}$ exists then we can cancel from products as

$$a^{-1}(a \cdot b) \equiv a^{-1}(a \cdot c) \pmod{n}$$

and so

$$(a^{-1}a) \cdot b \equiv (a^{-1}a) \cdot c \pmod{n}$$

and so

$$b \equiv c \pmod{n}.$$

Extended Euclidean algorithm and multiplicative inverses

Using linear combinations and the Euclidean algorithm we can show that

- for a in $\mathbb{Z}_n$, a multiplicative inverse of a modulo n will exists if and only if $\gcd(a, n) = 1$.
- **Terminology**: If $\gcd(a, b) = 1$ then a, b are said to be *relatively prime*, or *coprime*.

Extended Euclidean algorithm

- If $\gcd(a, b) = d$ then the *Extended Euclidean Algorithm* is an algorithm that produces integer coefficients x, y such that

$$xa + yb = d.$$

- It works by using the integer division with remainder equations from the Euclidean algorithm applied to $\gcd(a, b)$ and a method of *back substitution* to generate the x, y.
- We will demonstrate it by finding x, y so that $x \cdot 710 + y \cdot 310 = 10$.

Connection with multiplicative inverses

- When $\gcd(a, n) = 1$ and we find coefficients x, y such that

$$xa + yn = 1,$$

  then $x \equiv a^{-1} \pmod{n}$.

Extended Euclidean algorithm example

- From before, to find $\gcd(710, 310)$ we performed the following integer divisions.
- $710 = 2 \cdot 310 + 90$
- $310 = 3 \cdot 90 + 40$
- $90 = 2 \cdot 40 + 10$
- $40 = 4 \cdot 10 + 0$

To obtain a pair x, y that satisfy $x \cdot 710 + y \cdot 310 = 10$, we perform the following manipulations.

- $10 = 90 - 2 \cdot 40$, rearranging the second last division
- $10 = 90 - 2 \cdot (310 - 3 \cdot 90)$, replacing 40 with a rearrangment of the previous division
- $10 = -2 \cdot 310 + 7 \cdot 90$

In the last line we have expanded out the bracket, now we have 10 as a linear combination of 310 and 90. Continue workign backwards through the list of integer divisions from the Euclidean algorithm to get

- $10 = -2 \cdot 310 + 7 \cdot (710 - 2 \cdot 310)$, replacing the 90 using the first equation
- $10 = 7 \cdot 710 - 16 \cdot 310$, expanding out the bracket

We finally have what we want, the pair of coefficients, namely 7 and -16, that give a linear combination of 710 and 310 that is equal to their gcd.

- $\gcd(710, 310) = 10 = 7 \cdot 710 - 16 \cdot 310$

Prime numbers

Of central importance in cryptography, and of great interest to mathematicians, are the *prime integers*.

- **Definition - prime**

  And integer $p > 1$ is *prime* if its only positive divisors are 1 and p.

- **Definition - composite**

  A positive integer that is not prime is called a *composite* integer.

- The sequence of primes begins

$$p = 2, 3, 5, 7, 11, 13, 17, 19, 23, ...$$

- In fact, there are **infinitely many** primes. Known to Euclid, circa 2,300 years ago. See this Numberphile video for an accessible discussion of the proof of this, and its history.
- The largest prime known to humans is currently

$$2^{136,279,841} - 1,$$

  an integer with approximately 41 million digits. Discovered in Oct 2024, thanks to the GIMPS project.

Primes make integers

Primes are central to number theory thanks to

Theorem (Fundamental Theorem of Arithmetic) Every postive integer $n > 1$ can be written, uniquely, as a product of prime numbers,

$$n = p_1^{a_1} \cdot p_2^{a_2} \cdot \cdots \cdot p_r^{a_r},$$

where the p_i are primes and each a_i is a positive integer exponent.

- The expression in the theorem is known as the *prime factorization of* n and can be written compactly as

$$n = \prod_{i=1}^{r} p_i^{a_i}.$$

- The existence of prime factorizations follows immediately from the definition of a prime, i.e. *keep factoring* n *until you can factor no more.*
- The uniqueness part requires some careful mathematical argument.

Examples

- $91 = 7 \cdot 13$
- $3600 = 2^4 \cdot 3^2 \cdot 5^2$
- $11011 = 7 \cdot 11^2 \cdot 13$

Fermat's Little Theorem

We need to better understand the behaviour of *multiplication* and *exponentiation* on $\mathbb{Z}_n$ in order to understand how several *public key crytography* systems work. **Euler's Theorem** is a result that tells us a lot about how it behaves. A simpler first case to look at it called **Fermat's Little Theorem**.

Theorem (Fermat's Little Theorem) If p is a prime and a is a postive integer not divisible by p, (i.e. $a \not\equiv 0 \pmod{p}$) then

$$a^{p-1} \equiv 1 \pmod{p}.$$

- A proof for this is given in Stallings.

Example

With $a = 7$ and $p = 19$ we see

$$7^2 \equiv 49 \equiv 11 \pmod{19} \quad 7^4 \equiv (7^2)^2 \equiv 11^2 \equiv 121 \equiv 7 \pmod{19}$$

$$7^8 \equiv \left(7^4\right)^2 \equiv 7^2 \equiv 49 \equiv 11 \pmod{19} \quad 7^{16} \equiv \left(7^8\right)^2 \equiv 11^2 \equiv 121 \equiv 7 \pmod{19}$$

So now

$$a^{p-1} \equiv a^{18} \equiv a^{16+2} \equiv a^{16} \cdot a^2 \equiv 7^{16} \cdot 7^2 \equiv 7 \cdot 11 \equiv 77 \equiv 1 \pmod{19}.$$

- These calculations show an example of dealing with large exponents, (e.g. 18), by the method of **repeated squares**. More later.

Euler’s totient function

- Recall, the integer a has a multiplicative inverse modulo n if and only if $\gcd(a, n) = 1$, i.e. a and n are coprime (to each other).
- **Definition - Euler's totient function**

  Euler’s totient function $\phi : \mathbb{Z}^+ \to \mathbb{Z}^+$ is defined as: $\phi(n)$ is the number of positive integers a, less than n, (i.e. $1 \leq a < n$) such that $\gcd(a, n) = 1$.

Example

- $\phi(35) = 24$ as the integers coprime to 35 are

$$1, 2, 3, 4, 6, 8, 9, 11, 12, 13, 16, 17, 18,$$

$$19, 22, 23, 24, 26, 27, 29, 31, 32, 33, 34,$$

  and there are 24 integers on this list.
- Notice that $35 = 5 \cdot 7$ and this list omits all multiples of 5 and 7.
- This points to a more systematic way of evaluating $\phi(n)$.

Euler's totient function

Some evaluation formulae for ϕ are

- For a prime p,

$$\phi(p) = p - 1.$$

- For a power of a prime, p^a, we have

$$\phi(p^a) = p^{a-1}(p-1).$$

- ϕ is *multiplicative*, i.e.

$$\text{if } \gcd(a,b) = 1 \text{ then } \phi(a \cdot b) = \phi(a) \cdot \phi(b).$$

Euler's totient function

- Putting all these together, means that for an integer n with a prime factorization

$$n = \prod_{i=1}^{r} p_i^{a_i} = p_1^{a_1} \cdot p_2^{a_2} \cdot \cdots \cdot p_r^{a_r},$$

then

$$\phi(n) = \prod_{i=1}^{r} p_i^{a_i - 1}(p_i - 1) = p_1^{a_1 - 1} \cdot (p_1 - 1) \cdot p_2^{a_2 - 1} \cdot (p_2 - 1) \cdot \cdots \cdot p_r^{a_r - 1}(p_r - 1).$$

Example

$$\phi(35) = \phi(5 \cdot 7) = 5^0 \cdot 4 \cdot 7^0 \cdot 6 = 4 \cdot 6 = 24.$$

$$\phi(11011) = \phi(7 \cdot 11^2 \cdot 13) = 7^0 \cdot (7 - 1) \cdot 11 \cdot (11 - 1) \cdot 13^0 \cdot (13 - 1) = 6 \cdot 11 \cdot 10 \cdot 12 = 7920$$

Checking

```
1 import sympy as sp
2 n = 11011
3 L = [x for x in range(n) if sp.gcd(x,n)==1]
4
5 print(sp.factorint(n))
6 print(len(L))
7 print(sp.totient(n))
```



Evaluate

Euler's theorem

Finally, we can now state Euler's theorem, which is a generalization of Fermat's Little Theorem

Theorem (Euler's Theorem) If n is a postive integer modulus and a and n are coprime then

$$a^{\phi(n)} \equiv 1 \pmod{n}.$$

- A proof for this is given in Stallings.
- Theorem allows one to simplify powers of a modulo n, where the exponent is very large.
- Suppose that $b \equiv r \pmod{\phi(n)}$. Think of b being very large and r being relatively small.
- So $b = q \cdot \phi(n) + r$.
- Then

$$a^b = a^{q \cdot \phi(n) + r} = a^{q \cdot \phi(n)} \cdot a^r = \left(a^{\phi(n)}\right)^q \cdot a^r \equiv 1^q \cdot a^r \equiv a^r \pmod{n}.$$

Example

- Suppose $\gcd(a, 35) = 1$ and we want $a^{23458973249848} \pmod{35}$. Remember $\phi(35) = 24$.
- $23458973249848 \equiv 16 \pmod{24}$.
- So $a^{23458973249848} \equiv a^{16} \pmod{35}$.

Suggested reading

- During the week, read up on the following sections of Chapter 2 from Stallings.
    - Testing for Primality
    - Discrete Logarithms
- Discrete logarithms will be needed for public key encryption. I will cover it then also.
- A couple of appendix slides follow on the Chinese Remainder Theorem, also discussed in Chapter 2 of Stallings.
- Next week, our first encryption system, the *Data Encryption Standard* (DES).

The Chinese Remainder Theorem

- The CRT is another useful result for working with modular arithmetic.
- Suppose M is an integer factorized into *pairwise coprime* factors

$$M = \prod_{i=1}^{k} m_i = m_1 \cdot m_2 \cdot \ldots m_k,$$

  i.e. $\gcd(m_i, m_j) = 1$ for every pair of distint indices $1 \leq i, j, \leq k, i \neq j$.
- Such a factorization might be given by the different powers of primes in the prime factorization of M, i.e.

$$M = \prod_{i=1}^{k} p_i^{a_i} = (p_1^{a_1}) \cdot (p_2^{a_2}) \cdot \cdots \cdot (p_k^{a_k}) .$$

- The CRT describes a *one to one* mapping from the integers $\mathbb{Z}_M$ to the *Cartesian product*,

$$\mathbb{Z}_{m_1} \times \mathbb{Z}_{m_2} \times \cdots \times \mathbb{Z}_{m_k}.$$

- An element of $\mathbb{Z}_{m_1} \times \mathbb{Z}_{m_2} \times \cdots \times \mathbb{Z}_{m_k}$ is a k-tuple

$$(a_1, a_2, \ldots, a_k),$$

  where each a_i is a residue/remainder modulo m_i.

The Chinese Remainder Theorem

- $M = \prod_{i=1}^{k} m_i = m_1 \cdot m_2 \cdot \ldots m_k$,
- $\gcd(m_i, m_j) = 1$ for every pair of distint indices $1 \leq i, j, \leq k, i \neq j$.
- The mapping $\mathbb{Z}_M \to \mathbb{Z}_{m_1} \times \mathbb{Z}_{m_2} \times \cdots \times \mathbb{Z}_{m_k}$ is defined by

$$x \mapsto \Big((x \pmod{m_1}), (x \pmod{m_2}), \ldots, (x \pmod{m_k}) \Big)$$

  i.e. reduct x modulo m_i for the ith-component of the k-tuple.
- The mapping in the other direction $\mathbb{Z}_{m_1} \times \mathbb{Z}_{m_2} \times \cdots \times \mathbb{Z}_{m_k} \to \mathbb{Z}_M$ is a little more involved
- For each $1 \leq i \leq k$, define $M_i = M/m_i$, and let M_i^{-1} be the multiplicative inverse of M_i modulo m_i.
- Then, the k-tuple $(a_1, a_2, \ldots, a_k)$ will be mapped to the element a of $\mathbb{Z}_M$ defined by

$$a = \sum_{i=1}^{k} a_i \cdot M_i \cdot M_i^{-1} \pmod{M}.$$

- These two maps described above are *inverses* of one another.
- Arithmetic operations on elements of $\mathbb{Z}_M$ can be achived by corresepoding operations on elements of $\mathbb{Z}_{m_1} \times \mathbb{Z}_{m_2} \times \cdots \times \mathbb{Z}_{m_k}$.
- We will work with examples of this in the this week's lab.